

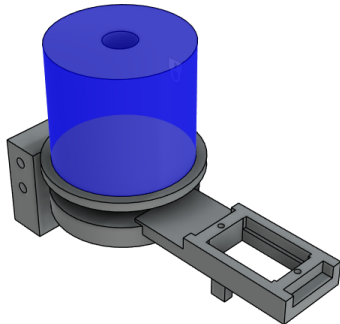
Revised head structural reference

Current T01 includes the tool-seat clearance slot; native solve completed and reviewed.

LOAD CASE AND MATERIAL

The outer **41.8 mm clamp cylinder** is fixed in X, Y and Z. The head datum annulus receives **+5 N in Z, +1 N in X** and a separate **-35 N mm moment about Y**. The current native report confirms the load components and fixed-face selection.

Isotropic solid PETG reference: **1,270 kg/m3** density, **2,000 MPa** Young's modulus, **0.38** Poisson ratio and **20 MPa** yield reference. These are modeling inputs; printed direction, walls, infill and filament strength remain unmeasured.



Current native fixed-face selection.

Source identity and original engineering records retained privately.

CURRENT NATIVE RESULTS

Quantity	Revised T01
Mesh input / order	0.75 mm / parabolic
Nodes / elements	1,027,487 / 724,705
Peak von Mises stress	0.279 MPa
Maximum displacement	0.0077216 mm
Minimum safety factor	71.662, native UI

The native geometry was matched to the revised assembly using exact BRep intersection and independently matching simulation working-model volume: **73,054.1741 mm3**. Source identity is retained privately. Native values use the displayed precision.

INTERPRETATION

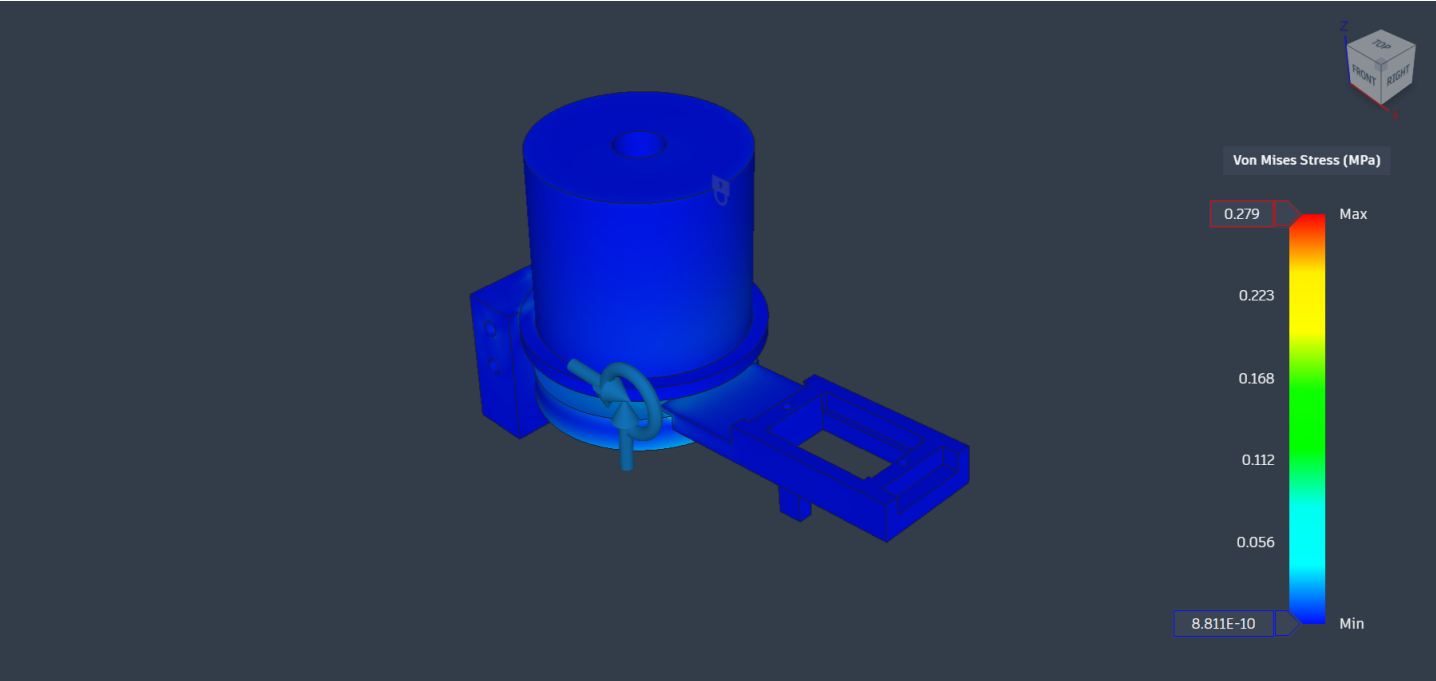
The result predicts approximately **7.72 micrometres** of maximum head deformation for this ideal reference case. It does not establish complete machine positioning accuracy, repeatability or the strength of an actual print. The large native safety factor applies only to the selected isotropic yield reference and boundary conditions.

The revised geometry has one refined solve. **No current peak-stress convergence pass is claimed.** Earlier mesh comparisons on the unrelieved head are historical evidence on page 3. Clamp slip, creep, fatigue, fastener preload, probe bending and physical commissioning remain outside this study.

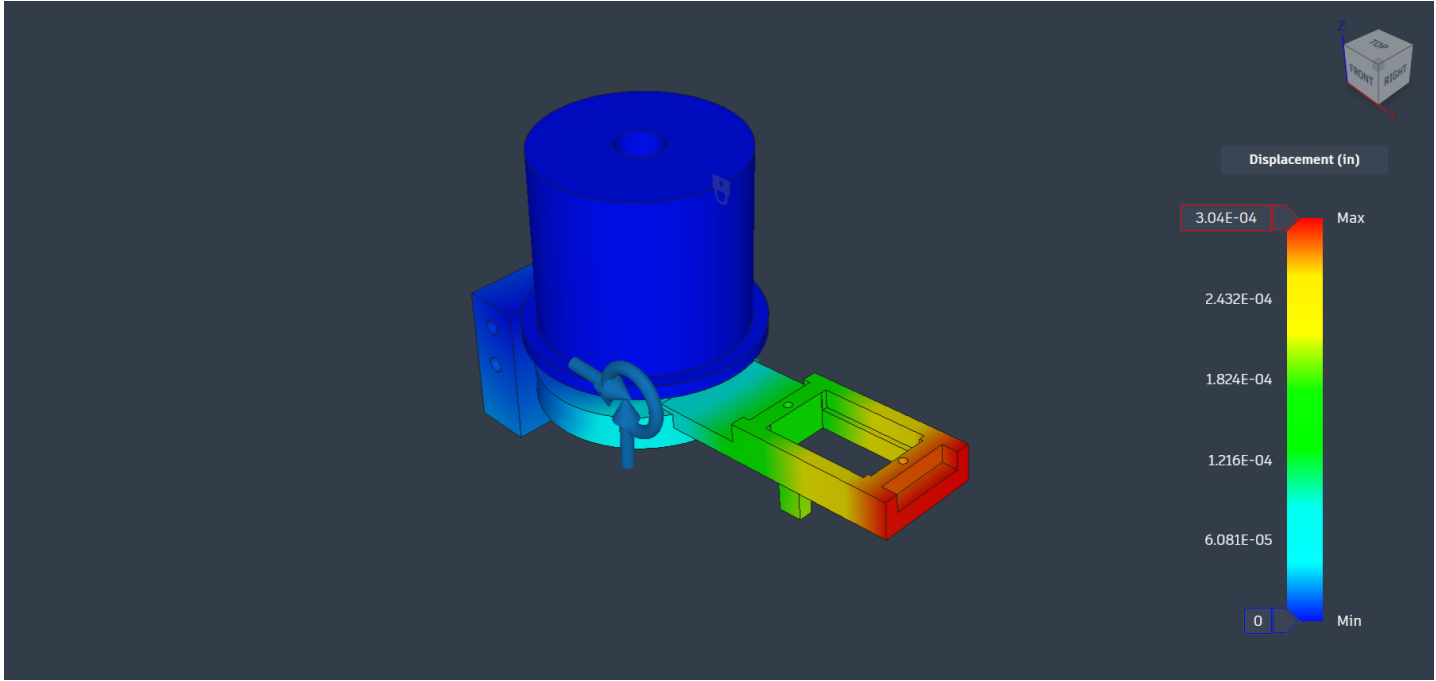
Current native result contours

Unaltered native report images from the revised T01 solve.

VON MISES STRESS / 0.279 MPa PEAK



TOTAL DISPLACEMENT / 0.0077216 mm MAX



The displacement legend uses inches: $3.04 \times 10^{-4} \text{ in} \times 25.4 = 0.0077216 \text{ mm}$. The contour colors indicate displacement magnitude; do not measure the displayed deformed shape. The native minimum safety factor was independently read as **71.662** from the current result-summary UI. It is consistent with the rounded stress and 20 MPa yield reference.

TRACEABLE NATIVE EVIDENCE

The editable study, native report and numerical records are retained privately. This public copy presents the reviewed results and their qualification limits.

Historical mesh comparison

Earlier unrelieved head geometry: retained for traceability, not current-revision qualification.

Quantity	1.00 mm mesh	0.75 mm mesh
Parabolic elements	325,921	745,530
Nodes	471,065	1,055,552
Peak von Mises stress	0.216 MPa	0.263 MPa
Maximum displacement	0.00718312 mm	0.00720852 mm
Minimum safety factor	92.527, native verified	Not independently verified

CONVERGENCE LIMIT

On the earlier geometry, displacement changed by **+0.354%** and global peak stress by **+21.76%** between these meshes. That supported stable displacement for the earlier fixed-clamp case. **Peak stress was not mesh-converged.** A sharp constraint boundary may contribute, but a stress singularity has not been established. The tool-seat relief changes the source geometry, so these historical runs are not a convergence series for the current head.

PHYSICAL AND SYSTEM LIMITS

The native full-assembly motion/contact checks and head linear-static solve address different questions. Neither is an energized hardware test. Purchased kit internals remain source-limited where manufacturer dimensions are unavailable. Measured fit, clamp grip, single-cord behavior, real print material response, electrical commissioning and motion control tuning require assembled hardware.

The 0.8 mm probe has a separate ideal beam screening calculation. Its calculated first-yield transverse load is about **0.72 N**; the head 1 N transverse load case does **not** qualify the thin probe for 1 N transverse tip loading.

Earlier native studies are retained privately for traceability; they do not qualify the current geometry.

Source identity and original engineering records retained privately.